A Discrete Model for the Performance of a Filter-Based Air Purifier in a Closed Room

Piyush Kumar

Abstract

A step-by-step model for how a filter-based purifier lowers pollutant levels in a closed room of nearly constant air volume. Pollutants are treated as a solute in the air (the solvent). The note sets out the model as written on the original handwritten page, states the range where it holds, gives a refinement, and compares both with the standard clean-air-delivery-rate (CADR) exponential decay in an illustrative simulation.

1 Setup and definitions

The room is closed (somewhat isolated) with nearly constant air volume. Pollutants are treated as solute particles in the air, which acts as solvent. Define

$$S_t = \frac{\text{amount of solute per unit volume}}{\text{amount of solvent per unit volume}} \quad \text{at time step } t.$$

Other quantities:

- Vol_t: volume passed through the filter at step t (starting just after $t-1$), with $\mathrm{Vol}_t = \mathrm{Vol}_0\, c(t-1)$, where $c(t)$ is the filter clog factor at time t;
- x_t: solvent per volume at time t;
- γ: an allowance factor that makes the filter realistic;
- Sol_0: amount of solute at time 0;
- α_t: amount of solute removed at step t.

2 The model

$$S_t = S_{t-1}\left(1 - \frac{\alpha_{t-1}}{\mathrm{Sol}_0 - (t-1)\,\alpha_0}\right), \qquad t \geq 1, \tag{1}$$

with $S_0 = S_{-1} = S_{-2} = \cdots$ by definition. The removal is

$$\alpha_0 \text{ measured at the start (a fixed input)}, \qquad \alpha_t = S_t\,\mathrm{Vol}_t\,x_t\,\gamma \quad (t \geq 1). \tag{2}$$

So the first-step removal α_0 is nonzero and measured, not computed; every later step follows the formula.

Reading the recurrence. The fraction $\alpha_{t-1}/(\cdot)$ is the share of the solute still present that the filter removes in one step, and S drops by that share. The denominator estimates the remaining solute by assuming each earlier step removed the same amount α_0.

Range of validity. The denominator reaches zero when $(t-1)\alpha_0 = \mathrm{Sol}_0$, so the model holds for

$$t < \frac{\mathrm{Sol}_0}{\alpha_0} + 1.$$

Since later steps remove less than α_0 (cleaner air, more clogging), the constant-α_0 estimate of what is left is too low, so the model decays somewhat faster than reality as t nears this limit. Early on it is accurate.

3 Refinement: exact remaining solute

Replacing the estimate with the solute actually left before step $t-1$,

$$S_t = S_{t-1}\left(1 - \frac{\alpha_{t-1}}{\mathrm{Sol}_0 - \sum_{k=0}^{t-2}\alpha_k}\right), \tag{3}$$

removes the singularity. (The sum has $t-1$ terms, matching the $(t-1)\alpha_0$ it replaces; if all $\alpha_k = \alpha_0$ the two agree.) If the remaining solute is proportional to S_{t-1}, the ratio becomes $\mathrm{Vol}_{t-1}x\gamma/(\text{room solvent})$, a per-step removal fraction that shrinks only through clogging.

4 Link to CADR decay

For a well-mixed room of volume V and a purifier with clean air delivery rate CADR, $C(t) = C_0 e^{-\mathrm{CADR}\,t/V}$. With no clogging ($c \equiv 1$) and constant x, the refinement removes a constant fraction f each step, so $S_t = S_0(1-f)^t \approx S_0 e^{-ft}$: the discrete CADR law with $f = \mathrm{CADR}\,\Delta t/V$. The clog factor c turns the constant rate into a slowly falling one, and γ accounts for imperfect capture.

5 Illustrative simulation

Parameters are **illustrative only, not measured**: $S_0 = 1$, $\mathrm{Sol}_0 = 100$, $\alpha_0 = 5$, $\mathrm{Vol}_0 = 10$, $x = 1$, $c(s) = e^{-0.02\max(s,0)}$, and $\gamma = 0.5$ (chosen so the formula would give α_0 at $t = 0$). The exponential uses $f = \alpha_0/\mathrm{Sol}_0 = 0.05$ with no clogging. Validity limit: $t < 21$.

step	model (1)	refinement (3)	$e^{-0.05t}$
$t = 10$	0.607	0.621	0.607
$t = 20$	0.243	0.420	0.368
$t = 60$	(outside range)	0.160	0.050

Up to $t \approx 10$ the model matches CADR decay closely. Near the limit it falls fastest, as expected. The refinement decays more slowly than the exponential because clogging reduces the flow.

6 Code

```
"""Illustrative simulation of Ray Parker's air purifier model.
ALL PARAMETERS ARE ILLUSTRATIVE, NOT MEASURED."""
import json, math, numpy as np, matplotlib; matplotlib.use("Agg")
import matplotlib.pyplot as plt
np.random.seed(0)  # fixed seed (model itself is deterministic)
```

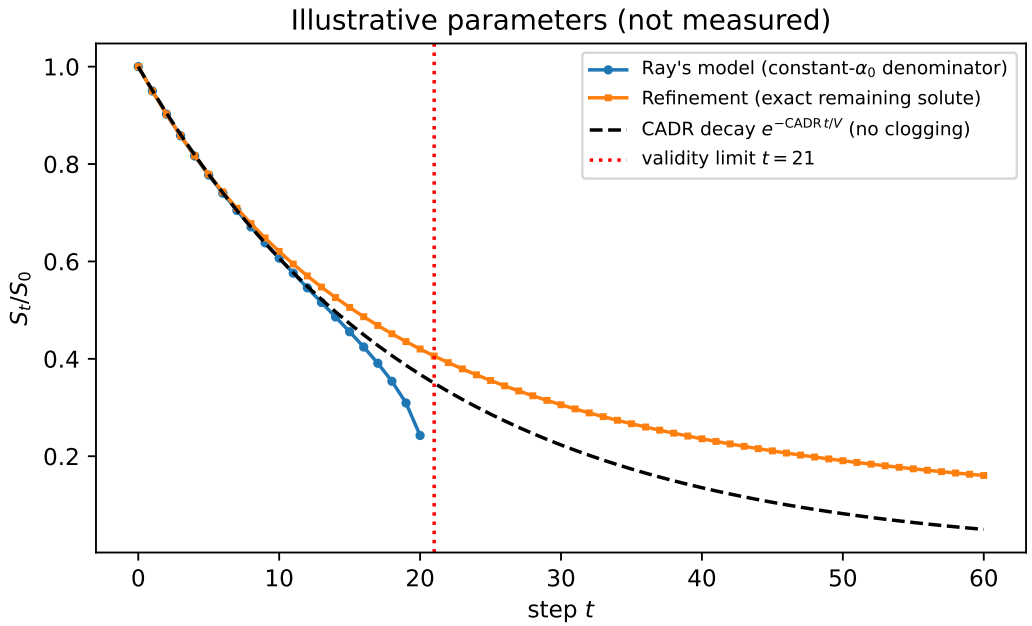
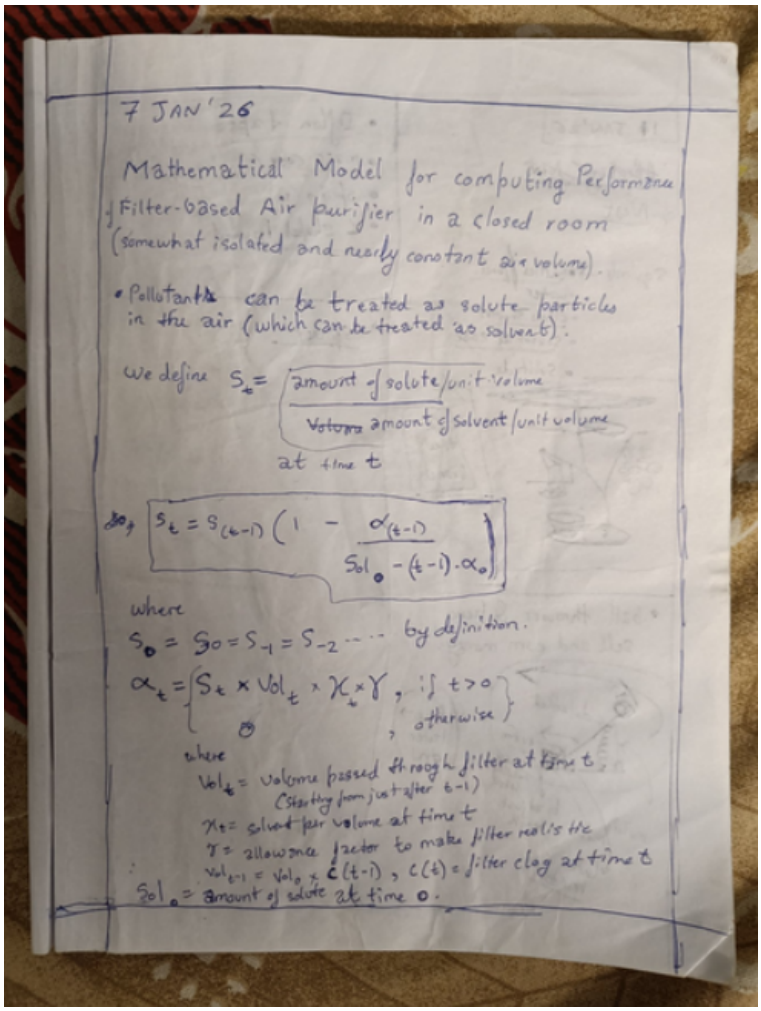


Figure 1: S_t/S_0 for the model, the refinement, and CADR decay (illustrative parameters).

```
S0, Sol0, alpha0 = 1.0, 100.0, 5.0      # alpha0: "measured" first-step removal (
    illustrative)
Vol0, x, k = 10.0, 1.0, 0.02            # volume per step, solvent per volume, clog
    rate
gamma = alpha0/(S0*Vol0*x)              # chosen so formula agrees with alpha0 at t
    =0
c = lambda s: math.exp(-k*max(s,0))     # clog factor, c(-1)=c(0)=1
Vol = lambda t: Vol0*c(t-1)
T = 60; tmax = Sol0/alpha0 + 1
def run(refined):
    S=[S0]; a=[alpha0]
    for t in range(1,T+1):
        den = Sol0-sum(a[:t-1]) if refined else Sol0-(t-1)*alpha0
        if den<=0: break
        S.append(max(S[-1]*(1-a[t-1]/den),0.0)); a.append(S[-1]*Vol(t)*x*gamma)
    return np.array(S)
Sr, Sf = run(False), run(True)
t = np.arange(T+1); Se = S0*np.exp(-(alpha0/Sol0)*t)   # CADR*dt/V = alpha0/Sol0
fig,ax=plt.subplots(figsize=(6.4,4))
ax.plot(range(len(Sr)),Sr,'o-',ms=3,label="Ray's␣model␣(constant-$\\alpha_0$␣
    denominator)")
ax.plot(range(len(Sf)),Sf,'s-',ms=2,label="Refinement␣(exact␣remaining␣solute)")
ax.plot(t,Se,'k--',label="CADR␣decay␣$e^{-\\mathrm{CADR}\\,t/V}$␣(no␣clogging)")
ax.axvline(tmax,color='r',ls=':',label=f"validity␣limit␣$t={tmax:.0f}$")
ax.set_xlabel("step␣$t$"); ax.set_ylabel("$S_t/S_0$"); ax.set_title("Illustrative␣
    parameters␣(not␣measured)")
ax.legend(fontsize=8); fig.tight_layout()
for e in ("png","pdf"): fig.savefig(f"simulation.{e}")
res=dict(params=dict(S0=S0,Sol0=Sol0,alpha0=alpha0,Vol0=Vol0,x=x,k=k,gamma=gamma),
  validity_limit=tmax, ray_steps=len(Sr)-1, S_ray_10=Sr[10], S_ref_10=Sf[10],
     S_exp_10=Se[10],
```

```
  S_ray_last=Sr[-1], S_ref_20=Sf[20], S_exp_20=Se[20], S_ref_60=Sf[60], S_exp_60=Se
      [60])
json.dump(res,open("results.json","w"),indent=1); print(res)
```

A Original handwritten page



The page is a rewritten copy made to explain the approach.